

Time allowed: 1 hour

All questions may be attempted.

Question 1

a) Find      (i)  $\int (5x-3)^{15} dx$  2

(ii)  $\int \frac{5}{\sqrt{3x+1}} dx$  3

b) If  $f(x) = \sqrt{\frac{2x-1}{4x+3}}$  then  $f'(x) = \frac{5}{\sqrt{(2x-1)(4x+3)^3}}$ . (You do not need to prove this!)

Use this fact to evaluate  $\int_{3/2}^{1/2} \frac{dx}{\sqrt{(2x-1)(4x+3)^3}}$  3

Question 2

a) Find the area enclosed by the curve  $y = 3 + 2x - x^2$ , the x-axis and the lines  $x = -1$  and  $x = 2$ . 3

b) Calculate the area between the curve  $y = x^3 - x^2 - 2x$  and the x-axis. 5

c) The area bounded by the curve  $y = x^3$ , the y-axis and the lines  $y = 0$  and  $y = k$  is 12 units. Find the value of  $k$ . 5

d) Find the area enclosed by the curve  $y = \frac{1}{4}x^3$  and the line  $y = x$ . 4

Question 3

Calculate the volume of the solid generated by rotating the ellipse  $\frac{x^2}{4} + y^2 = 1$  about the x-axis. 4

#### Question 4

Calculate an approximation for the area between the curve  $y = 2^x$ , the x-axis and the lines  $x = 1$ ,  $x = 3$

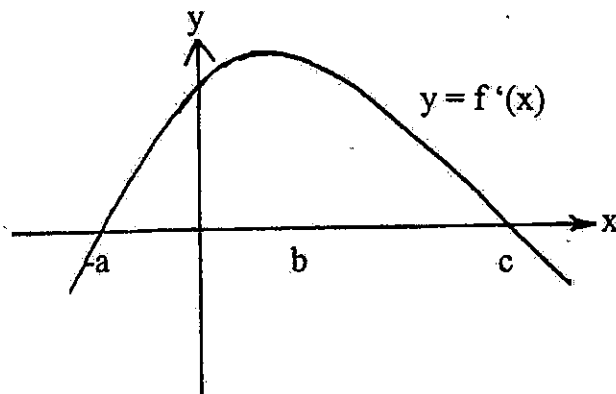
- a) by dividing the area into two strips and using the trapezoidal rule 2
- b) by using Simpson's rule with three function values. 2

#### Question 5

For the curve  $y = \frac{x^2}{x^2 - 4}$

- a) Show that there is a maximum turning point at  $(0,0)$ . 3
- b) Find, showing working, any
  - (i) vertical asymptotes 2
  - (ii) horizontal asymptotes. 2
- c) Is the function even or odd, give reasons? 1
- d) Draw a neat sketch of the function showing its main features. 2

#### Question 6



The parabola drawn in the diagram above shows the graph of the gradient function  $y = f'(x)$  for a given function  $y = f(x)$ . On separate diagrams draw possible graphs of

- a)  $y = f''(x)$  2
- b)  $y = f(x)$  3

### Question 1

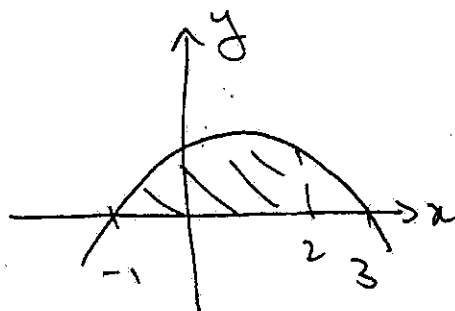
$$\begin{aligned} \text{a) (i)} \int (5x-3)^{15} dx &= \frac{1}{5} \times \frac{(5x-3)^{16}}{16} + C \\ &= \frac{(5x-3)^{16}}{80} + C \end{aligned}$$

$$\begin{aligned} \text{(ii)} \int \frac{5}{\sqrt{3x+1}} dx &= \int 5(3x+1)^{-\frac{1}{2}} dx \\ &= 5 \times \frac{(3x+1)^{\frac{1}{2}}}{\frac{1}{2}} \times \frac{1}{3} + C \\ &= \frac{10 \sqrt{3x+1}}{3} + C \end{aligned}$$

$$\begin{aligned} \text{b)} \int_{1\frac{1}{2}}^{5\frac{1}{2}} \frac{dx}{\sqrt{(2x-1)(4x+3)^3}} &= \frac{1}{5} \left[ \sqrt{\frac{2x-1}{4x+3}} \right]_{1\frac{1}{2}}^{5\frac{1}{2}} \\ &= \frac{1}{5} \left[ \sqrt{\frac{10}{25}} - \sqrt{\frac{2}{9}} \right] \\ &= \frac{1}{5} \left[ \frac{\sqrt{10}}{5} - \frac{\sqrt{2}}{3} \right] \end{aligned}$$

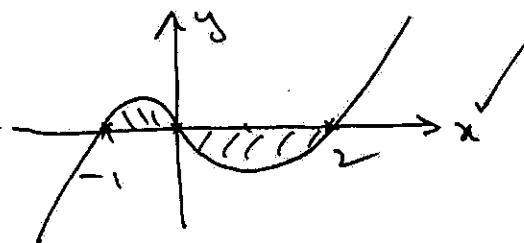
## Question 2

$$\begin{aligned} \text{a)} \quad & 3+2x-x^2=0 \\ & x^2-2x-3=0 \\ & (x+1)(x-3)=0 \\ & x=-1, 3 \end{aligned}$$



$$\begin{aligned} A &= \int_{-1}^3 (3+2x-x^2) dx \\ &= \left[ 3x + x^2 - \frac{x^3}{3} \right]_{-1}^3 \\ &= \left( 6 + 4 - \frac{8}{3} \right) - \left( -3 + 1 + \frac{1}{3} \right) \\ &= 9 \end{aligned}$$

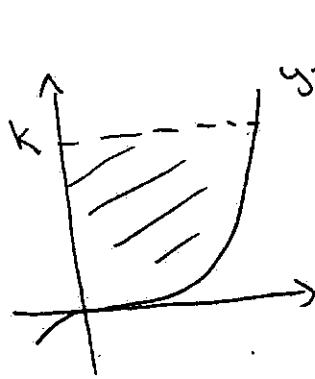
$$\begin{aligned} \text{b)} \quad & x(x^2-x-2)=0 \\ & x(x+1)(x-2)=0 \\ & x=0, -1, 2 \end{aligned}$$



$$\begin{aligned} A &= \int_{-1}^0 (x^3 - x^2 - 2x) dx + \left| \int_0^2 (x^3 - x^2 - 2x) dx \right| \\ &= \left[ \frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_{-1}^0 + \left| \left[ \frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_0^2 \right| \\ &= \left[ 0 - \left( \frac{1}{4} - \frac{1}{3} - 1 \right) \right] + \left| \left[ \left( 4 - \frac{8}{3} - 4 \right) - 0 \right] \right| \\ &= \frac{5}{12} + \left| \left( -\frac{8}{3} \right) \right| \\ &= \frac{37}{12} \end{aligned}$$

Q.2

c)



$$y = x^3 \Rightarrow x = y^{\frac{1}{3}}$$

$$\int_0^k y^{\frac{1}{3}} dy = 12 \checkmark$$

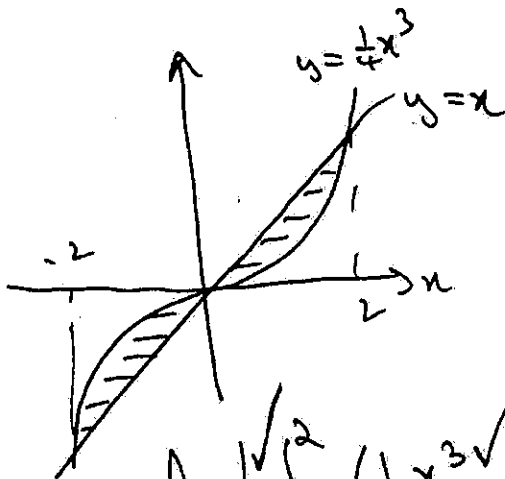
$$\left[ \frac{3}{4} y^{\frac{4}{3}} \right]_0^k = 12$$

$$\frac{3}{4} k^{\frac{4}{3}} = 12$$

$$k^{\frac{4}{3}} = 16 \checkmark$$

$$k = 8 \checkmark$$

d)



$$\frac{1}{4} x^3 = x$$

$$\frac{1}{4} x^3 - x = 0$$

$$x \left( \frac{1}{4} x^2 - 1 \right) = 0$$

$$x = 0, x = \pm 2 \checkmark$$

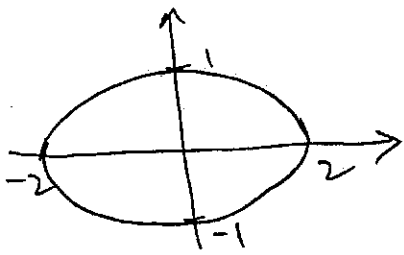
$$A = \left| \frac{1}{2} \int_{-2}^2 \left( \frac{1}{4} x^3 - x \right) dx \right|$$

$$= \left| \frac{1}{2} \left[ \frac{x^4}{16} - \frac{x^2}{2} \right]_{-2}^2 \right|$$

$$= \left| \frac{1}{2} [1 - 2] \right|$$

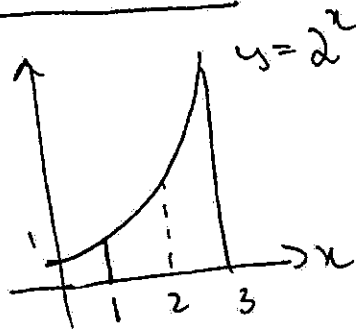
$$= 2 \checkmark$$

### Question 3



$$\begin{aligned}
 V &= \pi \int_{-2}^2 \left(1 - \frac{x^2}{4}\right) dx \quad \text{limits} \\
 &= 2\pi \int_0^2 \left(1 - \frac{x^2}{4}\right) dx \\
 &= 2\pi \left[ x - \frac{x^3}{12} \right]_0^2 \\
 &= 2\pi \left[ \left(2 - \frac{8}{12}\right) - 0 \right] \\
 &= 8\pi \frac{2}{3}
 \end{aligned}$$

### Question 4



Area trapez. ✓

$$\begin{aligned}
 a) A &\doteq \frac{1}{2} (2^1 + 2^3) + \frac{1}{2} (2^2 + 2^3) \\
 &= 9 \text{ u}^2 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 b) A &= \frac{3-1}{6} \{ 2^1 + 4 \times 2^2 + 2^3 \} \quad \checkmark \\
 &= \frac{1}{3} \{ 26 \} \\
 &= 8\frac{2}{3} \text{ u}^2 \quad \checkmark
 \end{aligned}$$

### Question 5

$$a) \quad y' = \frac{(x^2-4) \cdot 2x - x^2 \cdot 2x}{(x^2-4)^2}$$

$$= \frac{2x^3 - 8x - 2x^3}{(x^2-4)^2}$$

$$= \frac{-8x}{(x^2-4)^2} \quad \checkmark$$

$$y' = 0 \Rightarrow x = 0 \quad \checkmark$$

|      |                |   |                |
|------|----------------|---|----------------|
| $x$  | -1             | 0 | 1              |
| $y'$ | $+\frac{8}{9}$ | 0 | $-\frac{8}{9}$ |

$\checkmark$

$\therefore$  Max t. p. at  $(0,0)$

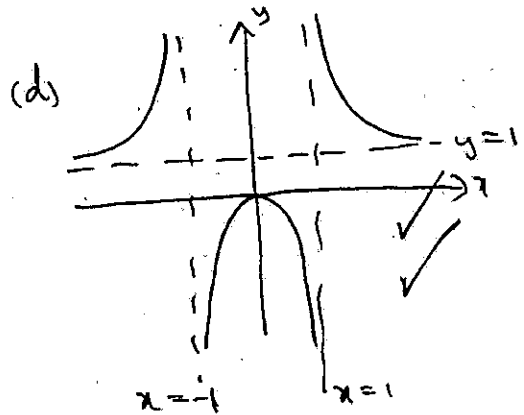
b) (i) Vert. asymptotes  
where  $x^2 - 4 = 0$   $\checkmark$   
ie.  $x = \pm 2$   $\checkmark$

$$(ii) \quad \lim_{x \rightarrow \pm \infty} \frac{x^2}{x^2-4} \checkmark = \lim_{x \rightarrow \pm \infty} \frac{1}{1-\frac{4}{x^2}}$$
$$= 1 \quad \checkmark$$

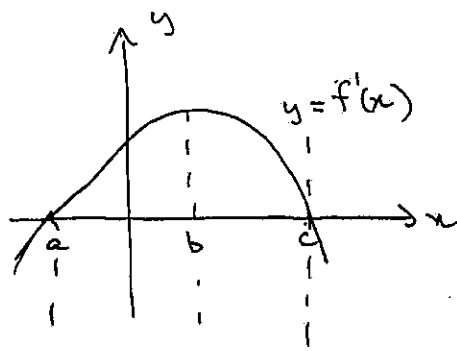
ie.  $y=1$  is horizontal asymptote

$$c) \quad f(-x) = \frac{(-x)^2}{(-x)^2-4}$$
$$= \frac{x^2}{x^2-4} = f(x) \quad \checkmark$$

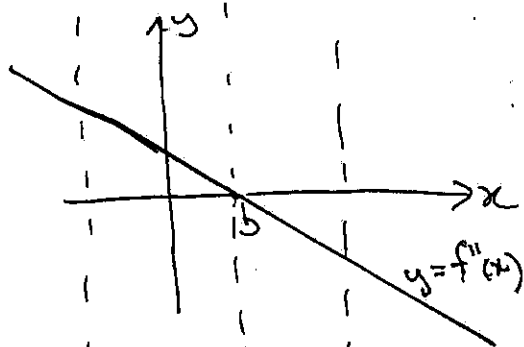
$\therefore y = f(x)$  is an even fn.



Q.6

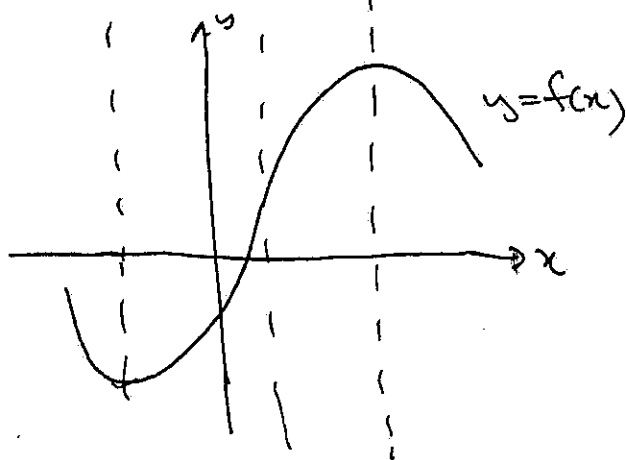


a)



St. Line correct slope ✓  
Through  $(b, 0)$  ✓

b)



Turning pts at  $x = a$  (Min) ✓  
 $x = b$  (Max) ✓

Inflection at  $x = b$  ✓